

Skills Summary

Common Denominators for Unlike Fractions

Use this reference while completing your worksheet or when studying.

Skill 1 — Rewriting a fraction with a new denominator

Rule: multiply the top and the bottom by the *same* number. The fraction keeps its value because you are multiplying by a form of 1. The smallest denominator that works for two fractions is the smallest number that both denominators divide into evenly.

Example: Rewrite $\frac{2}{3}$ with denominator 6.

Step 1. I want sixths and I have thirds — ask what number times 3 gives 6, then multiply top and bottom by it.

Step 2. $3 \times 2 = 6$, so multiply top and bottom by 2: $\frac{2 \times 2}{3 \times 2}$ — the new numerator is 4

Try it: Rewrite $\frac{3}{5}$ with denominator 10. What is the new numerator?

check: $\frac{6}{10}$

Skill 2 — Adding unlike fractions

Plan: you can only add pieces that are the same size. First rewrite both fractions with a common denominator, then add the numerators and keep the denominator. Simplify at the end.

Example: Add $\frac{1}{4} + \frac{1}{6}$.

Step 1. The pieces are different sizes — find a common denominator first. The smallest number both 4 and 6 divide into is 12.

Step 2. Rewrite: $\frac{1}{4} = \frac{3}{12}$ and $\frac{1}{6} = \frac{2}{12}$, so the sum is $\frac{3}{12} + \frac{2}{12} = \frac{5}{12}$

Try it: Add $\frac{1}{3} + \frac{1}{4}$.

check: $\frac{7}{12}$

Skill 3 — Subtracting unlike fractions

Plan: same idea as adding — rewrite both fractions with a common denominator, then subtract the numerators and keep the denominator.

Example: Subtract $\frac{3}{4} - \frac{2}{3}$.

Step 1. Different denominators — rewrite both in twelfths, since 12 is the smallest number 4 and 3 both divide into.

Step 2. $\frac{3}{4} = \frac{9}{12}$ and $\frac{2}{3} = \frac{8}{12}$, so the difference is $\frac{9}{12} - \frac{8}{12} = \frac{1}{12}$

Try it: Subtract $\frac{7}{8} - \frac{1}{2}$.

check: $\frac{3}{8}$

(!) Watch out: never add or subtract the denominators. In $\frac{1}{4} + \frac{1}{6}$ the answer is *not* $\frac{2}{10}$ — the denominator tells the size of the pieces, and only same-size pieces can be counted together.